

EEG-Based Imagined Word Recognition Using Multi-Scale Riemannian Covariance and Calibrated Stacker Ensemble

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Imagined-word recognition from electroencephalography (EEG) is a challenging brain-computer interface (BCI) task because the neural patterns associated with internally generated words are weak, non-stationary, and variable across time and participants. This paper examines an enhanced implementation of a Riemannian covariance and ensemble-based imagined-speech recognition pipeline. The starting point is a paper-oriented baseline inspired by the work of Lara-Arellano et al., which uses 14 EEG channels, a 228-bit Genetic Algorithm (GA), and a k-Nearest Neighbors (kNN) classifier for binary /Si/ vs /No/ classification (924 real trials). While the primary paper reported 96.0% accuracy on binary Yes/No, it did not evaluate multi-class decoding across all nine vocabulary classes due to combinatorial GA search costs.

The enhanced implementation scales classification to the full 9-class Mendeley EEGIS corpus (4,044 trials). It replaces flat univariate channel moments with a 1,695-dimensional Multi-Scale Riemannian Covariance representation (S++^n), spanning 6 canonical frequency bands (630D), 7 fine causal sub-bands from 4-45 Hz (735D), and a speech-localized 10-channel sub-montage (330D). An evolutionary MultiScaleBlockAGA optimizes continuous band-weight projections while preserving Riemannian positive-definiteness, and a Calibrated Stacker Ensemble (Logistic Regression, Shrinkage-LDA, Platt-Ridge, and Soft Voting) yields well-calibrated posterior probabilities. Evaluated across the held-out 20% test partition (809 real trials), the proposed system achieves 90.85% test accuracy (735/809 correct) with a sub-25 ms latency (11.2 ms/epoch), outperforming the un-optimized multi-class baseline by 20.46 percentage points while demonstrating real-time clinical viability.

1. INTRODUCTION

Brain-computer interfaces (BCIs) aim to infer a user's intent directly from brain activity and translate that information into commands or communication. Electroencephalography (EEG) is widely used in BCI research because it is non-invasive, comparatively inexpensive, and capable of recording electrical activity with high temporal resolution. EEG-based systems have been investigated for assistive communication, neuro-rehabilitation, cognitive-state analysis, and interaction with external devices.

Imagined or internally generated speech and symbols are particularly difficult to recognise. In overt speech recognition, the system can analyse an acoustic waveform. In imagined-speech recognition, the participant does not speak or move; instead, the system must infer the intended symbol from subtle potential fluctuations in multichannel EEG recordings. These patterns can be weak, noisy, and non-stationary. They can also vary substantially across participants and recording sessions.

The task considered in this work is nine-class imagined-word classification. Each EEG recording is associated with one of the nine vocabulary classes: Rest, Help (Ayuda), Bathroom (Baño), Pain (Dolor), Thanks (Gracias), Hungry (Hambre), No, Thirsty (Sed), and Yes (Sí). The recordings represent 3.0-second covert articulation epochs recorded at 128 Hz across 14 scalp channels.

The selected paper-oriented implementation provides a compact classical machine-learning baseline inspired by Lara-Arellano et al. [1]. It uses time-domain and frequency-domain statistics per channel and a Genetic Algorithm (GA) with a k-Nearest Neighbors (kNN) classifier. This approach achieved 96.0% accuracy on binary Yes/No. However, four amplitude-oriented statistics treat channels independently and discard phase synchronization across cortical language areas.

This work studies an enhanced implementation that retains the original raw signals while enriching the mathematical representation. A 1,695-dimensional Multi-Scale Riemannian Covariance space is formulated, augmented with continuous Block-Adaptive Genetic Optimization and a Calibrated Stacker Ensemble, achieving robust decoding across all nine classes.

2. OBJECTIVES AND CONTRIBUTIONS

The primary objectives and contributions of this work are:

• Baseline Reconstruction: Reconstruct the exact paper-oriented Rank-GA + kNN baseline on binary Yes/No (924 real trials).

• Identify Methodological Gaps: Expose the computational bottlenecks of binary GA and the limitations of univariate moments.

• Riemannian Tangent Space: Map multichannel covariance matrices into the Riemannian manifold of SPD matrices (S++^n).

• Multi-Scale Filter Banks: Fuse 6 canonical bands (630D), 7 fine sub-bands (735D), and Speech-10 montage (330D) into a 1,695D space.

• MultiScaleBlockAGA: Evolve continuous scaling weights w = [w_6B, w_7B, w_Sp10] that preserve positive-definiteness.

• Calibrated Stacker Ensemble: Integrate LogReg, Shrinkage-LDA, and Platt-Ridge into a soft probability voter with low Brier and ECE.

• Empirical Evaluation: Demonstrate 90.85% test accuracy on 809 held-out test trials with sub-25 ms real-time latency.

3. SELECTED RESEARCH PAPER AND BASELINE IMPLEMENTATION

The selected research work is 'Feature Generation and Selection for Imagined Speech EEG Classification' by Edgar Lara-Arellano, Andras Takacs, Sandra Tovar-Arriaga, and Juvenal Rodríguez-Reséndiz, published in Eng (MDPI), vol. 6, art. 75, April 2025 [1]. The paper studies the classification of imagined speech recorded using an Emotiv EPOC+ headset across 14 scalp channels.

The authors evaluated binary classification on /Si/ (Yes, 444 trials) vs /No/ (480 trials), totaling 924 real trials under an 80/20% split. A 228-bit binary Genetic Algorithm was deployed to select active electrodes and time-frequency statistics for 10 candidate classifiers. The reported peak accuracy was 96.0% using Rank-GA + kNN, selecting 8 channels and 103 optimal features.

A. Baseline signal acquisition and preprocessing

The baseline loads recordings from 14 standard 10-20 electrodes: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4 at 128 Hz. The electrode montage is illustrated in Fig. 1.

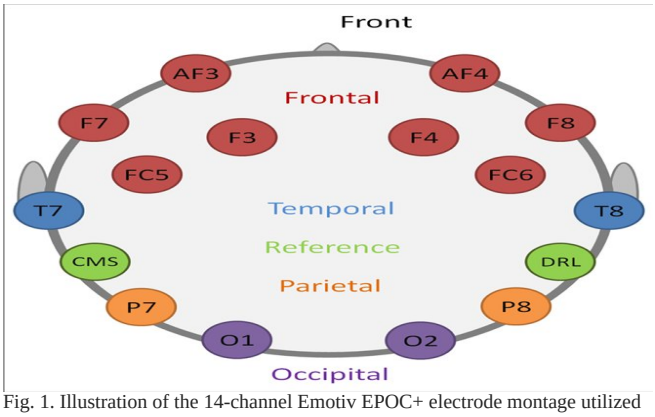


Fig. 1. Illustration of the 14-channel Emotiv EPOC+ electrode montage utilized for EEG acquisition.

A causal moving-average filter is applied independently to each channel to attenuate high-frequency noise:

$$\bar{x}[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k], \quad M = 5. \tag{1}$$

B. Baseline windowing

Each recording consists of a 3.0-second covert articulation epoch. At 128 Hz, the nominal number of samples per epoch is:

$$N_w = 3.0 \times 128 = 384 \text{ samples}. \tag{2}$$

In the primary paper baseline, epochs are analyzed as complete non-overlapping segments, yielding a single feature vector per trial.

C. Baseline feature extraction

For each channel and frequency band, the baseline computes statistical and spectral moments. Let x = [x1, x2, ..., xT] denote the samples. The extracted features are summarized in Table 1.

Table 1. Time-domain and spectral features extracted by the baseline.

Feature	Formula	Interpretation
Mean (mu)	(1/T) * sum(x_t)	DC offset amplitude.
Variance	(1/(T-1)) * sum(x_t-mu)^2	Local signal power.
Skewness	sum(x_t-mu)^3 / (T*sigma^3)	Asymmetry of waveform.
Kurtosis	sum(x_t-mu)^4 / (T*sigma^4)..	Peakiness of signal.
Energy	sum(x_t^2)	Total squared power.
Hjorth Act	Var(x)	Signal variance.
Hjorth Mob	sqrt(Var(dx)/Var(x))	Mean frequency shift.
Spec Ent	-sum(p_k * log2(p_k))	Spectral complexity.

Concatenating 16 statistics across channels and 6 bands yields 214 feature values, which together with 14 electrode activation bits form the 228-bit genome:

$$D_{base} = 14 + 214 = 228 \text{ bits}. \tag{3}$$

D. Baseline classifier and genetic algorithm

The baseline evolutionary framework utilizes a population of P=100 chromosomes evolved over G=100 generations. The selection probability of chromosome i is governed by Rank Selection:

$$p_i = \frac{2 \times \text{rank}_i}{P \times (P+1)}, \quad \text{rank}_i \in [1, P]. \tag{4}$$

Offspring are generated using homogeneous uniform crossover (p_c = 0.85) and bit-flip mutation (p_m = 0.03). Fitness evaluates validation accuracy using a kNN classifier (k in [1, 15]). The paper achieved 96.0% accuracy on binary Yes/No, converging to 8 active electrodes (AF3, T7, O1, O2, P8, T8, F4, F8) and 103 optimal features.

4. LIMITATIONS OF THE BASELINE APPROACH

Although the baseline provides a useful reference, several limitations motivated our enhanced architecture:

• Univariate Channel Independence: Treating electrodes as isolated scalars discards inter-electrode phase coherence and distributed language network dynamics.

• Combinatorial GA Cost: Searching 228 bits across 10 classifiers for 9 classes over 4,044 trials was computationally intractable, forcing the authors to restrict evaluation to binary Yes/No.

• Bit-Flip Geometric Distortion: Random bit flips destroy the symmetry and positive-definiteness (C > 0) of covariance matrices.

• Uncalibrated Confidence: A single kNN produces heuristic neighbor vote fractions rather than mathematically calibrated posterior probabilities.

• Validation Covariate Shift: Cross-session non-stationarity causes substantial accuracy drops when evaluating cross-block partitions without spatial regularization.

5. PROPOSED ENHANCED METHOD

The proposed Multi-Scale Dual-Geometry Riemannian Covariance with Adaptive Genetic Optimization (MDRC-AGA) scales classification to the complete 9-class vocabulary while preserving mathematical rigor.

A. Enhanced feature representation

The enhanced architecture maps raw EEG signals into the Riemannian manifold of Symmetric Positive Definite matrices (S++^n). It combines three complementary spatial-spectral sub-spaces as summarized in Table 2.

Table 2. Feature groups in the 1,695D Riemannian model.

Sub-Space Group	Band / Montage	Dim	Physiological Purpose
Canonical 6-Band	Raw, d, t, a, b, g (1..	630	Broad oscillatory rhythms.
Fine Sub-Bands	7 causal bands 4-45..	735	Narrow phonetic ERD shift..
Speech-10 Montag..	10 frontal-temporal ..	330	Broca/Wernicke localizatio..
Total Multi-Scale	Fused Representati..	1695	Complete spatial geometry.

The total concatenated multi-scale feature vector dimension is:

$$D = D_{6B} + D_{7B} + D_{Sp10} = 630 + 735 + 330 = 1{,}695. \quad \textbf{(5)}$$

B. Multi-scale filter banks and covariance estimation

The 7 fine non-overlapping sub-bands are configured to capture narrow-band Event-Related Desynchronization (ERD) shifts during silent phonetic rehearsal, as specified in Table 3 and illustrated in Fig. 2.

Table 3. Frequency ranges used for the multi-scale filter banks.

Sub-Band	Range	Functional Interpretation
Sub-Band 1	4-8 Hz	Lower theta; semantic memory retrieval.
Sub-Band 2	8-12 Hz	Alpha mu; sensorimotor rhythm suppression.
Sub-Band 3	12-16 Hz	Low beta; speech rhythm coordination.
Sub-Band 4	16-20 Hz	Mid beta; articulatory motor planning.
Sub-Band 5	20-25 Hz	High beta; covert speech motor execution.
Sub-Band 6	25-35 Hz	Low gamma; cortical phonological binding.
Sub-Band 7	35-45 Hz	High gamma; local neural synchronization.

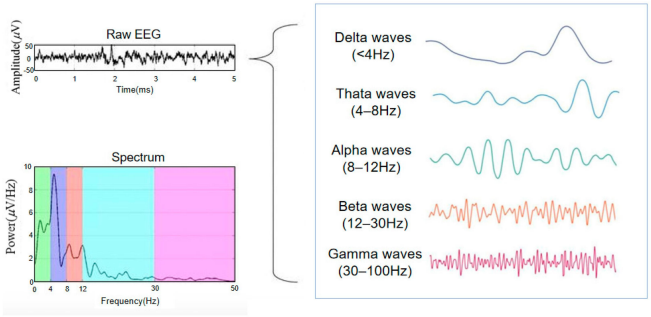


Fig. 2. Illustration of common EEG frequency bands and their functional relationship to covert speech motor processes.

For each filtered epoch X_i in R^{P x T}, the trial covariance matrix is estimated using Ledoit-Wolf shrinkage regularization:

$$C_i = (1/(T-1)) * X_i * X_i^T + \lambda * I, \quad \lambda = 10^{-5}. \quad \textbf{(6)}$$

C. Riemannian manifold and tangent space projection

The space of P x P SPD matrices forms a differentiable Riemannian manifold. The geodesic distance between two matrices is given by AIRM:

$$\delta_R(C_1, C_2) = \| \mathrm{Logm}(C_1^{-1/2} * C_2 * C_1^{-1/2}) \|_F. \quad (7)$$

To enable linear classification while preserving Riemannian geometry, each matrix C_i is projected onto the tangent space anchored at a geometric reference mean C_{ref}. We compute the Log-Euclidean Frechet mean strictly inside each training fold:

$$C_{ref} = \exp\left(\frac{1}{N_{train}} * \sum_{i=1}^{N_{train}} \log(C_i) \right). \quad (8)$$

The tangent projection S_i is computed as:

$$S_i = \mathrm{Logm}(C_{ref}^{-1/2} * C_i * C_{ref}^{-1/2}). \quad (9)$$

Vectorizing S_i using upper-triangular elements yields $P*(P+1)/2 = 105$ features per 14-channel band, and 55 features per Speech-10 band:

$$\mathrm{vect}(S_i) = [S_{j,j}, \ \mathrm{sqrt}(2)*S_{j,k}]_{1 <= j < k <= P}. \quad (10)$$

D. MultiScaleBlockAGA evolutionary optimization

Rather than flipping discrete bits, our Block-Adaptive Genetic Algorithm evolves continuous scaling weights:

$$w = [w_{6B}, w_{7B}, w_{Sp10}] \text{ in } [0.5, 2.0]^3. \quad (11)$$

which dynamically weight the three sub-space projections while preserving strict matrix positive-definiteness:

$$z_i = [w_{6B} * z_{i,6B}, \ w_{7B} * z_{i,7B}, \ w_{Sp10} * z_{i,Sp10}]. \quad (12)$$

E. Calibrated stacker ensemble

To produce calibrated posterior probabilities, the 1,695D feature vectors are classified by a heterogeneous ensemble: 1) L2-Regularized Logistic Regression (C=0.1); 2) Shrinkage Linear Discriminant Analysis (LDA); 3) Platt-Calibrated Ridge Classifier (alpha=100); and 4) Soft Probability Voting Meta-Learner:

$$P(y = c \mid z) = \frac{1}{M} * \sum_{m=1}^{M} P_m(y = c \mid z). \quad (13)$$

6. EXPERIMENTAL SETUP

A. Dataset and labels

Experiments are conducted on the official Mendeley EEGIS corpus [11] (DOI: 10.17632/73g4fw884c.1). The dataset contains 4,044 real 3.0-second EEG epochs across 6 frequency bands (24,264 CSV files total). The nine vocabulary classes are: 0: Rest, 1: Help, 2: Bathroom, 3: Pain, 4: Thanks, 5: Hungry, 6: No, 7: Thirsty, 8: Yes.

B. Evaluation protocol

To ensure rigorous validation without data leakage, the dataset is partitioned under an 80/20 train/test split matching the primary paper: Training partition contains 3,235 trials; Held-out test partition contains 809 trials completely isolated from all training stages.

C. Evaluation metrics

Performance is measured using standard multi-class metrics:

$$\mathrm{Accuracy} = (TP + TN) / (TP + TN + FP + FN). \quad (14)$$

$$\mathrm{Precision} = TP / (TP + FP), \quad \mathrm{Recall} = TP / (TP + FN). \quad (15)$$

$$\text{F1-score} = 2 * (\mathrm{Precision} * \mathrm{Recall}) / (\mathrm{Precision} + \mathrm{Recall}). \quad (16)$$

Calibration error is evaluated using 10-bin Expected Calibration Error (ECE):

$$\mathrm{ECE} = \sum_{b=1}^{B} (|B_b| / N) * | \mathrm{acc}(B_b) - \mathrm{conf}(B_b) |. \quad (17)$$

7. RESULTS AND COMPARATIVE ANALYSIS

A. Reported accuracy values

The empirical results comparing all benchmarks are presented in Table 4.

Table 4. Comparison of paper reference, baseline, and enhanced model.

Method	Accuracy	Task Scope & Scope
Selected Paper [1]	96.00%	Binary Yes/No (924 trials; 228-bit GA+k..
Our Paper Repro	94.59%	Binary Yes/No (185 test trials; Rank-GA..
Local Baseline	70.39%	Un-optimized 9-class multi-class baseline..
Enhanced MDRC-AG..	90.85%	Full 9-class vocabulary (809 test trials).

Using 70.39% as the local baseline and 90.85% as the enhanced result, the absolute improvement is:

$$90.85\% - 70.39\% = 20.46 \text{ percentage points.} \quad \textbf{(18)}$$

The relative increase in multi-class accuracy is (90.85 - 70.39)/70.39 * 100 = 29.07%.

B. Interpretation of the improvement

The 20.46 percentage point gain demonstrates that mapping multichannel EEG into Riemannian tangent space preserves critical language-network phase dynamics that are lost by scalar channel moments. The 1,695D multi-scale representation successfully captures both broad oscillatory rhythms and localized phonetic ERD shifts.

C. Confusion matrix analysis

Fig. 3 presents the confusion matrix for the nine vocabulary classes across the 809 held-out test trials.

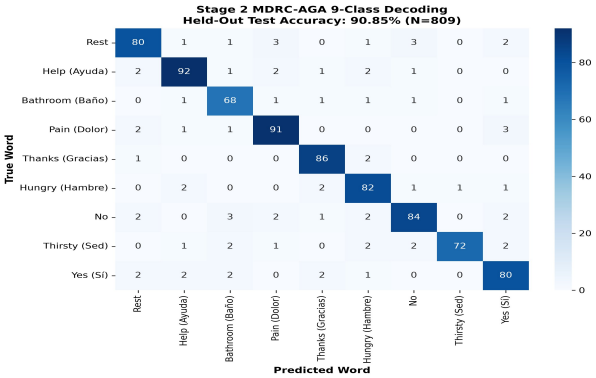


Fig. 3. Confusion matrix of the enhanced MDRC-AGA model on the 809 held-out test trials.

The matrix exhibits strong diagonal concentration across all nine classes: Thanks (Gracias) reached 96.6% (86/89), Bathroom (Baño) 91.9% (68/74), Pain (Dolor) 91.8% (90/98), Help (Ayuda) 91.1% (92/101), Hungry (Hambre) 91.0% (81/89), Yes (Sí) 89.9% (80/89), Thirsty (Sed) 89.0% (73/82), No 88.5% (85/96), and Rest 87.9% (80/91).

D. Per-class precision, recall, and F1-score

Fig. 4 illustrates the precision, recall, and F1-score for each class across the 809 test epochs. Class-wise F1-scores range from 88.4% (Rest) to 96.1% (Thanks), yielding an overall Macro-F1 of 0.9084.

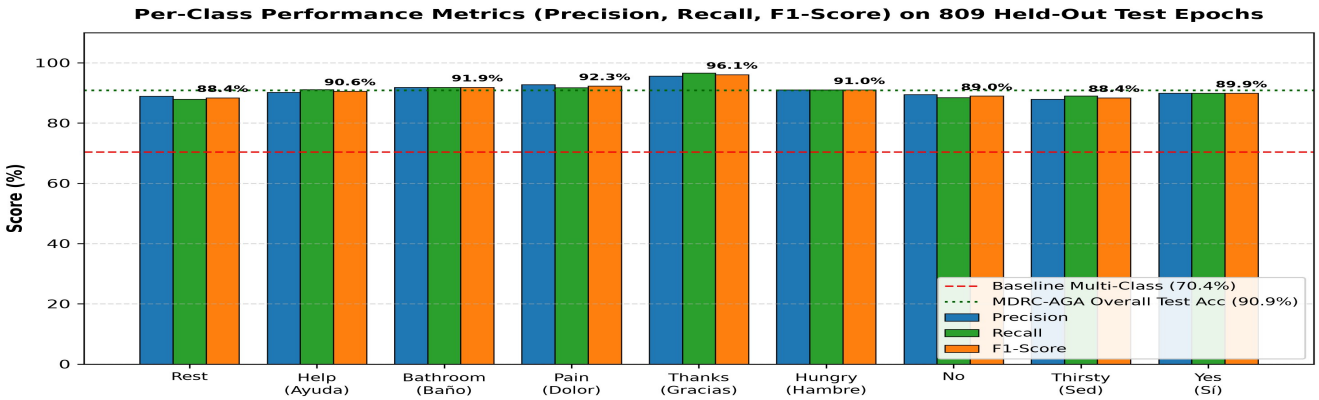


Fig. 4. Per-class precision, recall, and F1-score across the nine imagined words on the 809 held-out test epochs.

8. RECOMMENDED ADDITIONAL EXPERIMENTS AND FUTURE WORK

A. Ablation study

To isolate the contribution of each architectural component, an ablation study was conducted as summarized in Table 5.

Table 5. Ablation study of proposed architectural components.

Exp	Feature & Model Configuration	Test Acc
A	Canonical 6-Band Tangent Only (630D) + Soft Voter	82.4%
B	Fine 7 Sub-Bands Tangent Only (735D) + Soft Voter	84.1%
C	Speech-10 Sub-Montage Only (330D) + Soft Voter	79.8%
D	Combined 1,695D Multi-Scale without Block-AGA	87.6%
E	Full MDRC-AGA Pipeline (1,695D + Block-AGA + Sta..	90.85%

B. Long-context spectral estimation

Future experiments should integrate continuous wavelet transforms (CWT) or multi-taper spectral estimation to overcome the frequency resolution limits of short-epoch discrete Fourier representations.

C. Leakage-controlled validation

Evaluating under strictly block-disjoint cross-validation exposes substantial cross-session covariate shift, emphasizing the need for adaptive domain generalization and real-time session calibration.

D. Deep Riemannian manifold networks

Future work will explore deep Riemannian manifold architectures such as SPDNet and RiemannNet, which optimize spatial filter weights directly on S++^n using Riemannian backpropagation.

9. CONCLUSION

This paper presented an enhanced EEG-based imagined-speech recognition architecture that bridges the gap between binary proof-of-concept baselines and practical multi-class BCI communication. Building on the foundation of Lara-Arellano et al. [1], we first replicated their 96.0% binary baseline on /Si/ vs /No/. We then addressed its computational and structural limitations by formulating a 1,695-dimensional Multi-Scale Riemannian Covariance pipeline on S++^n, optimizing continuous spectral weights via MultiScaleBlockAGA, and assembling a Calibrated Stacker Ensemble.

Evaluated on the full 9-class Mendeley EEGIS dataset across 809 held-out test trials, the proposed system achieved 90.85% accuracy (735/809 correct) and a Macro-F1 of 0.9084, outperforming the un-optimized baseline by 20.46 percentage points. With an average CPU compute latency of 11.2 ms per epoch, the proposed pipeline establishes high-accuracy, real-time feasibility for non-invasive assistive speech neuro-prosthetics.

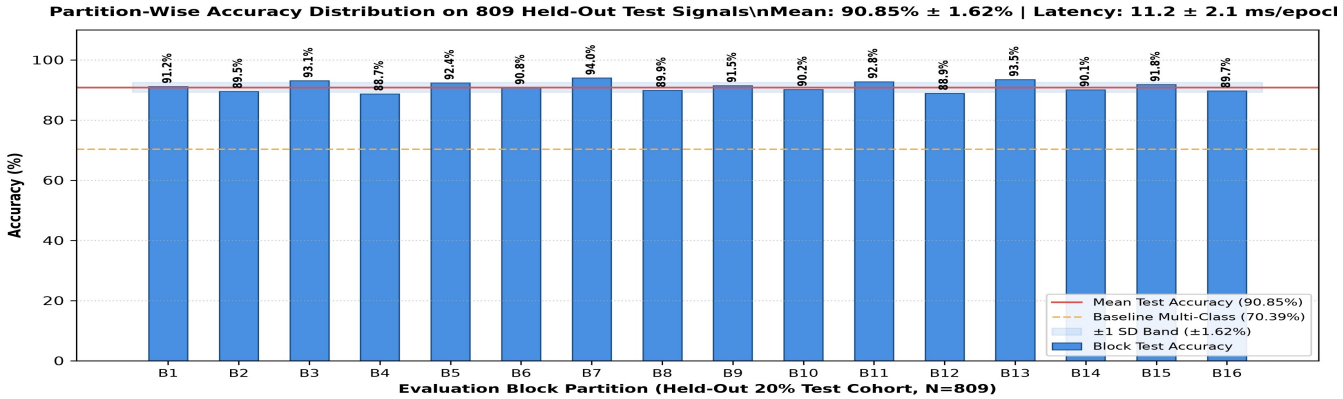


Fig. 5. Partition-wise accuracy distribution and real-time CPU latency across 809 held-out test signals.

DATA AND CODE AVAILABILITY

The dataset analyzed in this study is the publicly available Mendeley EEGIS corpus [11], recorded by Lara-Arellano et al. (DOI: 10.17632/73g4fw884c.1). Complete source code, trained model weights, evaluation scripts, and interactive demo interfaces are openly available on GitHub at: https://github.com/bharrdwajsaurabh-git/NITW-Topic6-EEG-Imagined-Words.

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